the cold plate, where the heat transfer cycle repeats.

Solid-state heat pumps. These semiconductor devices are often referred to as thermoelectric coolers (TECs). These are thin and compact, and typically placed between a heat source and a heat-sink to facilitate quicker heat dissipation. When a voltage is applied to a TEC, a temperature differential is formed between the two sides of the device. This enables heat transfer to occur through conduction.

While TECs are not very efficient, these still move substantial amounts of heat and have a very long service life. TECs are ideal for applications requiring spot cooling or where sub-ambient temperatures are required.

In addition, if current is reversed on a TEC, heat transfer flow reverses. This essentially turns the device into a heater. This can be ideal for applications where precise temperature control is needed.

Electrostatic fluid acceleration. An electrostatic fluid accelerator (EFA) pumps a fluid, such as air, without any moving parts. Instead of using rotating blades, as in a conventional fan, an EFA uses an electric field to propel electrically-charged air molecules. Because air molecules are normally neutrally charged, the EFA has to create some charged molecules, or ions, first.

Thus, there are three basic steps in the fluid acceleration process: ionising air molecules, using those ions to push many more neutral molecules in a desired direction and then recapturing and neutralising the ions to eliminate any net charge.

The basic principle has been understood for some time but only in recent years has seen developments in the design and manufacture of EFA devices that may allow these to find practical and economical applications, such as in micro-cooling of electronic components.

Role of nanotechnology. For the first time, an international team of scientists led by a researcher at University of California, Riverside has modified the energy spectrum of acoustic phonons—elemental excitations, also referred to as quasi-particles, that spread heat through crystalline materials like a wave—by confining these to nanometre-scale semiconductor structures.

The results have important implications in the thermal management of electronic devices. The team used semiconductor nanowires from gallium arsenide (GaAs), synthesized by researchers in Finland, and an imaging technique called Brillouin-Mandelstam light scattering spectroscopy (BMS) to study the movement of phonons through crystalline nanostructures. By changing the size and the shape of GaAs nanostructures, the researchers were able to alter the energy spectrum, or dispersion, of acoustic phonons.

Controlling phonon dispersion is crucial for improving heat removal from nano-scale electronic devices, which has become a major roadblock in allowing engineers to continue to reduce their size. It can also be used to improve the efficiency of thermoelectric energy generation. In this case, decreasing thermal conductivity by phonons is beneficial for thermoelectric devices. EFY

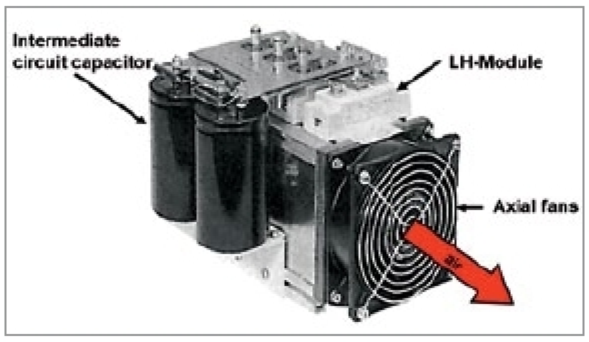

Fig. 5: Forced air cooling of electronics (Credit: www.powerguru.org)

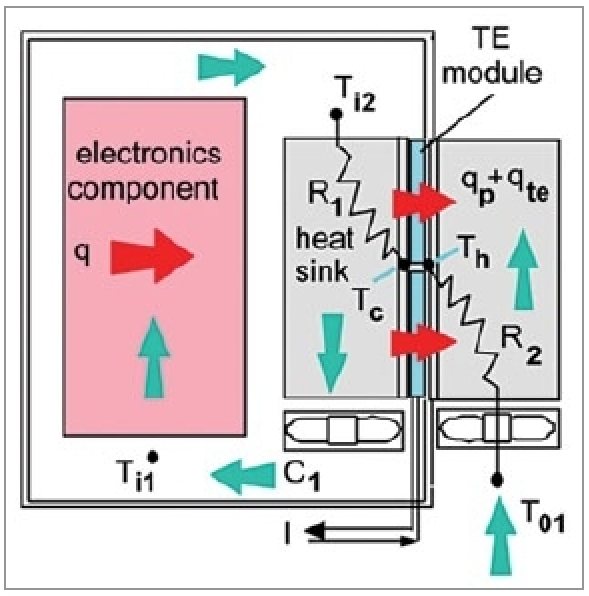

Fig. 6: Thermoelectric cooling (Credit: www.electronics-cooling.com)